

10. Iodomethane, CH_3I , reacts with a range of nucleophiles. The rate of its reaction with thiosulfate ions, $\text{S}_2\text{O}_3^{2-}$, has been studied at a range of temperatures.

(a) The reaction is found to be first order with respect to iodomethane and has a rate constant of $3.10 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ at a particular temperature.

(i) Write the rate equation for this reaction. [2]

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(ii) The activation energy for this reaction is 81.1 kJ mol^{-1} with a frequency factor, A , of $5.62 \times 10^{12} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$. Find the temperature used for this reaction. [3]

$T = \text{..... K}$



(b) One method of performing this experiment involves sampling and quenching.

(i) Explain what is meant by the term 'quenching' and why it needs to be carried out. [2]

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(ii) One method of measuring the concentration of thiosulfate ions in solution is by titration using a solution containing iodine. Suggest an indicator for this titration. [1]

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(iii) A student suggests using gravimetric analysis to measure the amount of iodide ions produced in the reaction by addition of $\text{Pb}^{2+}(\text{aq})$ ions to form a precipitate. Give the colour of this precipitate. [1]

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(c) Iodomethane is an intermediate in the conversion of methanol to ethanoic acid.

(i) One catalyst used for this reaction is the iridium-containing complex ion $[\text{Ir}(\text{CO})_2\text{I}_2]^-$. It is an example of a homogeneous catalyst.

I. State what is meant by a *homogeneous* catalyst. [1]

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II. Explain why transition metal complexes such as $[\text{Ir}(\text{CO})_2\text{I}_2]^-$ can act as homogeneous catalysts. [2]

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- (ii) Ethanoic acid is a weak acid with $K_a = 1.76 \times 10^{-5} \text{ mol dm}^{-3}$. Calculate the pH of an aqueous solution of ethanoic acid with a concentration of $0.220 \text{ mol dm}^{-3}$. [3]

pH =

Examiner
only

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